

Bungee Drop

(Coach Scheduled Event - 45 Minutes)



Description: Teams will design and construct a bungee cord system to drop a weight from a designated height. Teams will also answer questions on a written assessment on the principles of the law of conservation of energy, elastic potential energy, gravitational potential energy, and kinetic energy.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Teams may also bring:

- Writing utensils (must bring for the written assessment)
- Ruler or meter stick
- Scissors

Materials Provided at the Event:

- An assortment of rubber bands, sizes #64, #33, #19, and #117B.
- 50 cm (19.5 in) of 2.54 cm (1 in) masking tape

Bottle Weight provided by the ES for the drops (pictured):

- One 16.9 oz water bottle attached to a carabiner
- Solid material, such as BBs or sand, will be put in the water bottle so that the entire bottle assembly (including the carabiner) will mass 100 g.

Water Bottle



THE COMPETITION

Written Assessment: The team will answer assessment questions simultaneously with the 25-minute building phase on the principles of the law of conservation of energy, elastic potential energy, gravitational potential energy, and kinetic energy.

Construction Phase (25 min):

- The Event Supervisor will announce the designated height, the Target Drop Distance (TDD), to the artificial horizon at the start of the construction phase. The Target Drop Distance is measured from the bottom of the Bottle Weight when the weight is hung by the carabiner from the drop apparatus. The Target Drop Distance can be from 200 cm to 500 cm.

- At the bottom of the drop, a paper chart will have a horizontal line that is the artificial horizon. This line is the zero point, with additional horizontal lines above it and below it marked from -100 cm (below) to +100 cm (above).
- The team will choose and connect rubber bands to create a bungee cord. They may cut the rubber bands to any size or leave them whole and may tie or tape them together in any manner they choose.

Bungee Testing:

- Each team will have 2 drops for this competition.
- For each drop, the partners will:
 - Clip the water bottle weight to a rubber band on their bungee. The team may attach the water bottle at the very end of the bungee or at a rubber band higher up.
 - Hang the other end of their bungee to the Retaining Hook on the Bungee Drop Platform, making sure that the rest of the rubber bands on their bungee are hanging loose over the front of the platform.
 - Hang the water bottle's carabiner from the drop release mechanism.
 - They will announce to the Event Supervisor they are ready to drop and wait for a reply from the supervisor.
- The Event Supervisor will set their video recording device in such a way as to show the entire Artificial Horizon Chart and record the fall in slow motion.
 - Once the supervisor has the camera ready, the team will trigger the release mechanism to drop the water bottle.
 - The recording must have visible in the background a horizon chart marked in 1 centimeter increments as the distance gauge, visible in the background, +100 cm to -100 cm.
- The Event Supervisor will review the recording and record the distance from the horizon, plus or minus, to the nearest centimeter mark before the team makes the second drop. The distance from the horizon will be measured to the bottom of the water bottle.
- The team may use the information from the first drop to alter where on the bungee they hang the water bottle for the second drop. They may not alter the bungee itself between drops.

SCORING: Final Score (FS) = Assessment Score + Best Drop Score, max score = 100 points

- Assessment score = $(TAS \div CAS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = team's score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Drop Score = $(CDS \div TDS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Competition Drop Score (CDS) = the best Team Drop score over all drops
 - Team Drop Score (TDS) = Difference x Possible Penalty
 - Difference = absolute value of the distance from the 0 horizon
 - If the bottom of the bottle does not fall within 100 cm on either side of the horizon, the distance from the horizon will be recorded as 200 cm.
 - Possible Penalty: If the bottom of the bottle falls past the horizon, the Difference will be multiplied by 3.
 - If the bungee breaks or the weight falls off the bungee on the team's first drop, the team will receive a Drop Score of 0 and will not receive a second drop. If either of these issues happens on the second drop, the team will receive the score from their first drop.

Scoring Example

- Competition Assessment Score (CAS) = 9.
- The team scores 6 on the assessment (TAS).
 - Assessment Score for the team = $(6 \div 9) \times 50 = 33.33$ points
- Competition Drop Score (CDS) = 5 cm
- Team's best drop distance = - 6 cm (bottle bottom passes the 0 horizon by six centimeters)

- Best Team Distance Score = absolute value(-6) x 3 = 18
 - Drop Score = $(5 \div 18) * 50 = 13.89$
- Final Score (FS) = $33.33 + 13.89 = 47.22$

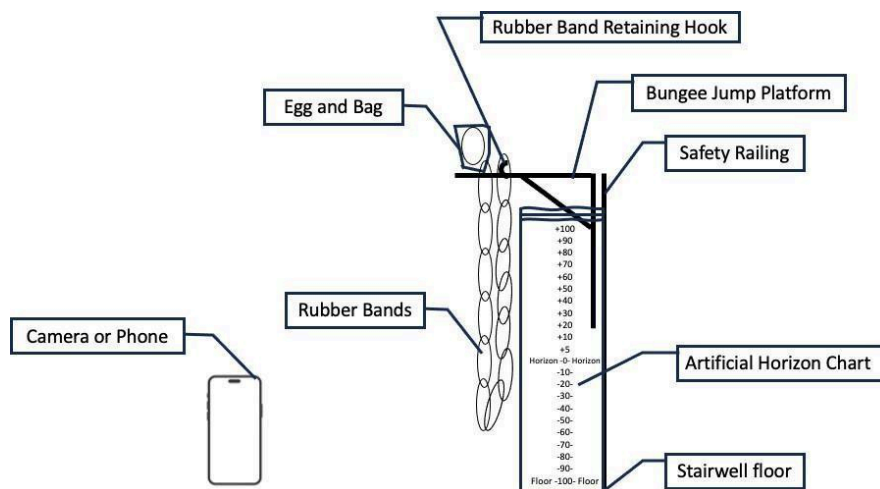
Tiebreakers

1. Written Assessment Score
2. Correct answers on selected written assessment questions

Resources

- [Bungee-Jumping Egg – Flinn Scientific \(.pdf download\)](#)
- [Egg Bungee Teacher Notes – Science Scope, February 2005 \(.pdf download\)](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Competition Diagram (diagram is from previous year. A water bottle weight replaces the egg.)



Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles